

Course Reminders

- Due Sunday: D5, A3, Q6, *Weekly project survey (EC)*

Notes:

- D3 scores released
- Issues created: who your graders are & rubrics

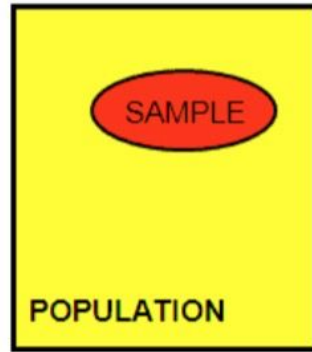
Nonparametric Statistics

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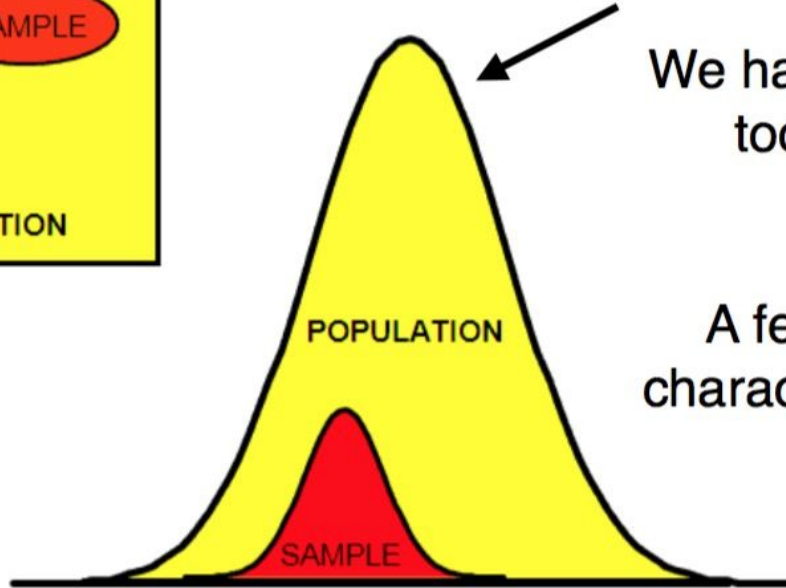
Non-parametric Statistics: The Why



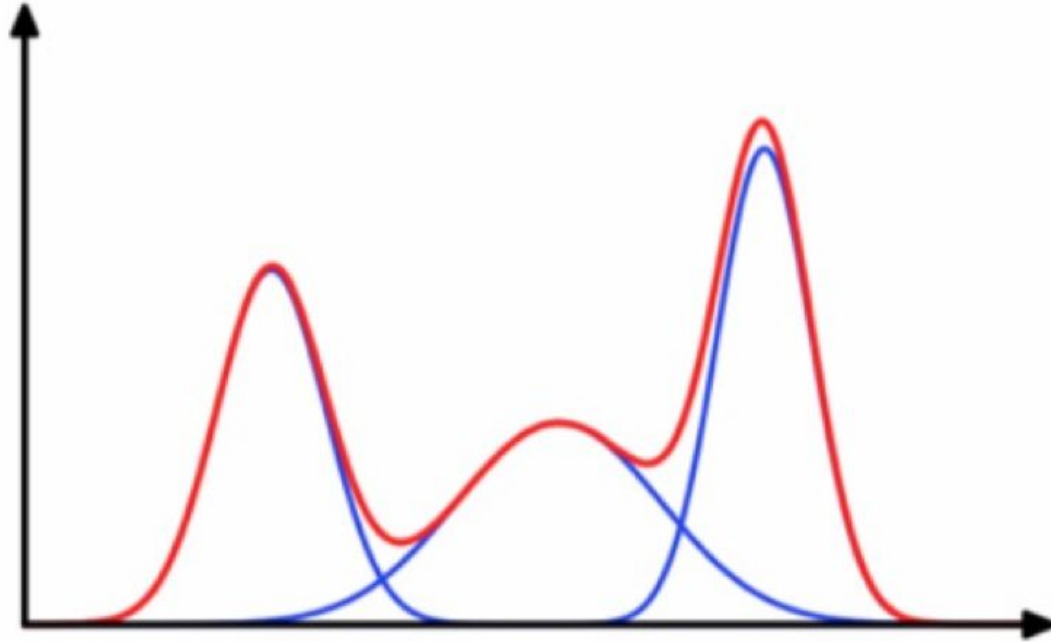
Normal distribution
(nice and friendly)

We have good math
tools for this.

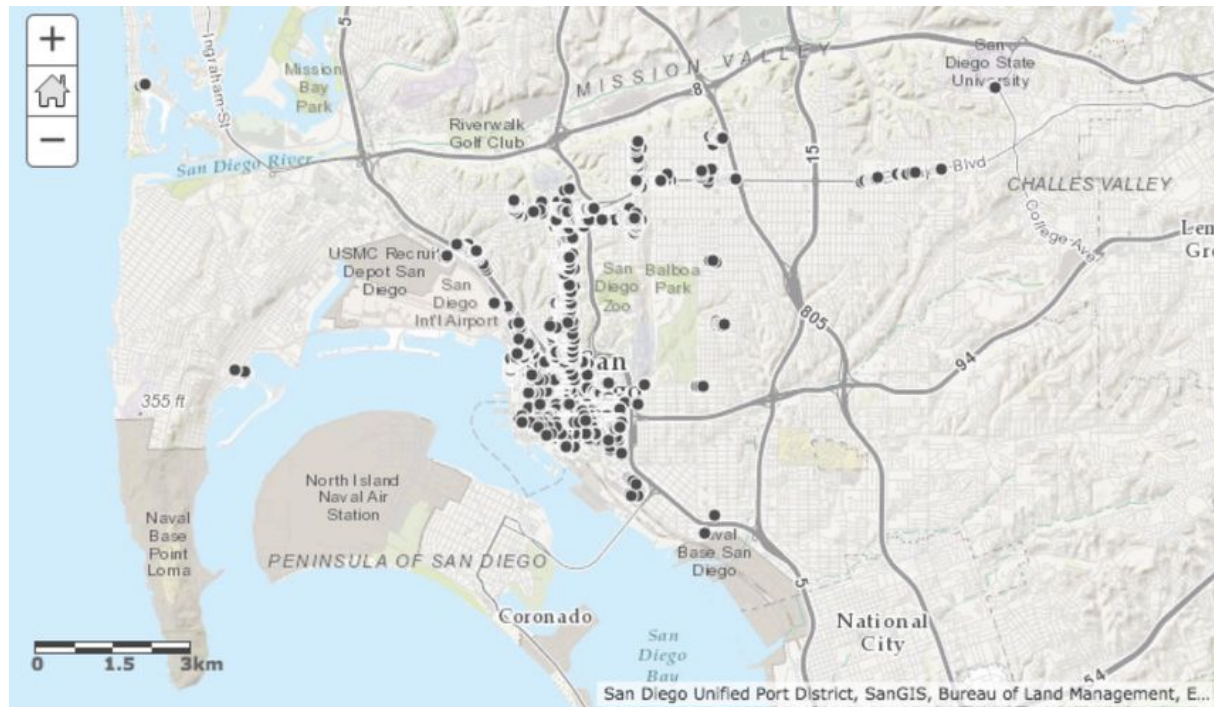
A few parameters **fully**
characterize the distribution.



Non-parametric Statistics:
What if your distribution looks like this?



Non-parametric Statistics: ...or like this?

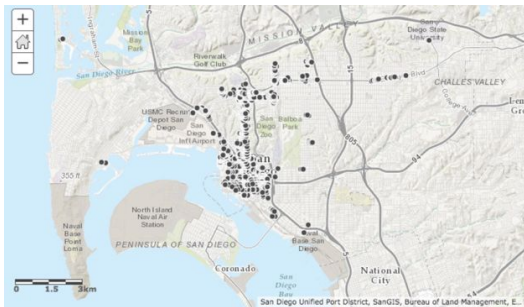


Parameters (like mean and variance) cannot fully and accurately capture this distribution!

Hence, we require **non-parametric statistics**.

When to turn to non-parametric statistics...

- When underlying distributions are non-normal, skewed, or cannot be parameterized simply.



- When you have ranked (ordinal) data, *e.g.*, preferences.

Like	Like Somewhat	Neutral	Dislike Somewhat	Dislike
1	2	3	4	5

- When you need to build an empirical “null” distribution.

Non-parametric Statistics: distribution-free

- **Myth:** Non-parametric statistics does not use parameters.
- **Fact:** Non-parametric statistics does not make *assumptions about* / parametrize the underlying distribution generating the data.

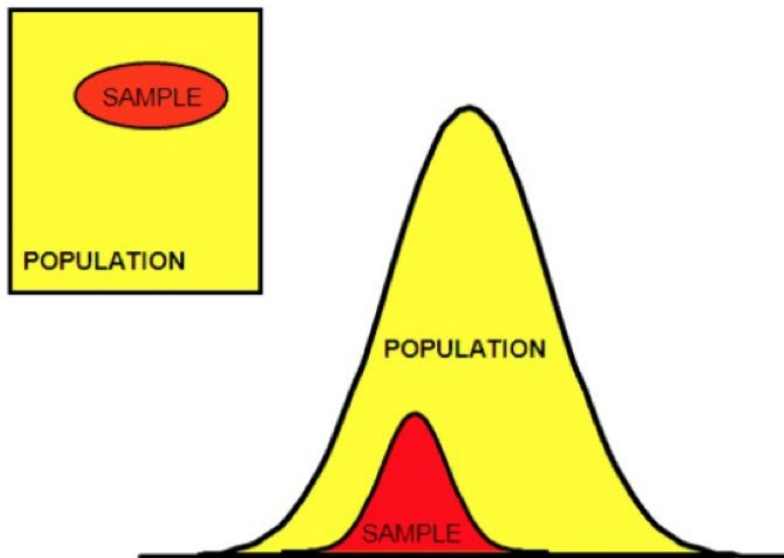
- **“Distribution-Free” statistics**
 - Meaning, it does not assume data-generating process (like heights) result in, *e.g.*, normally-distributed data

Resampling statistics: The What

- Bootstrap (Monte Carlo)
- Rank Statistics (Mann Whitney U)
- Kolmogorov-Smirnoff Test
- Non-parametric prediction models

1) Bootstrapping (resampling)

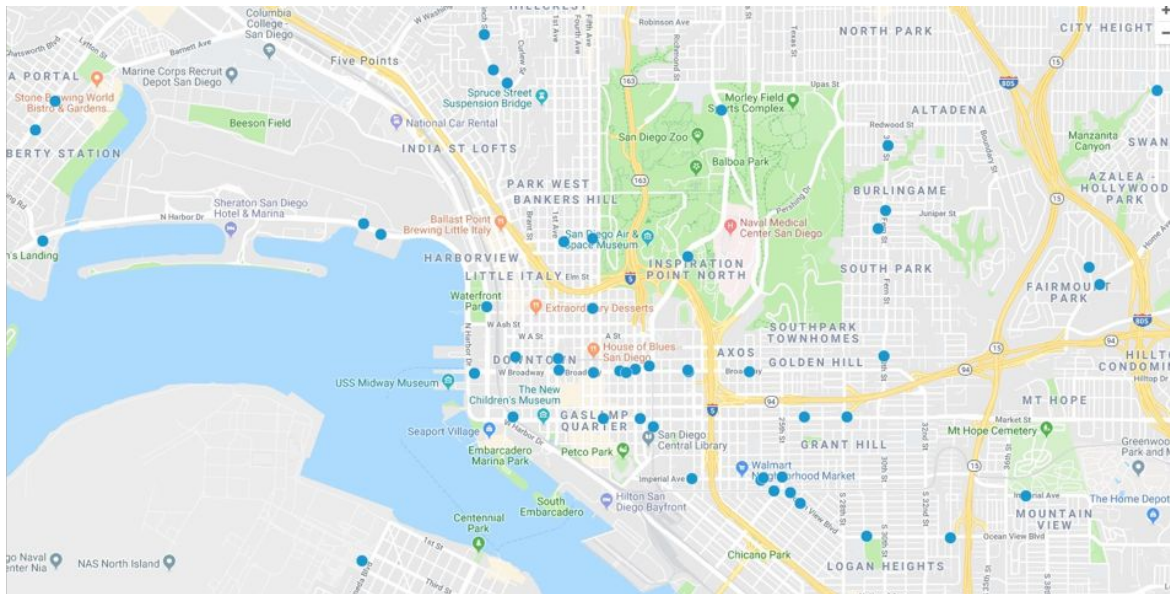
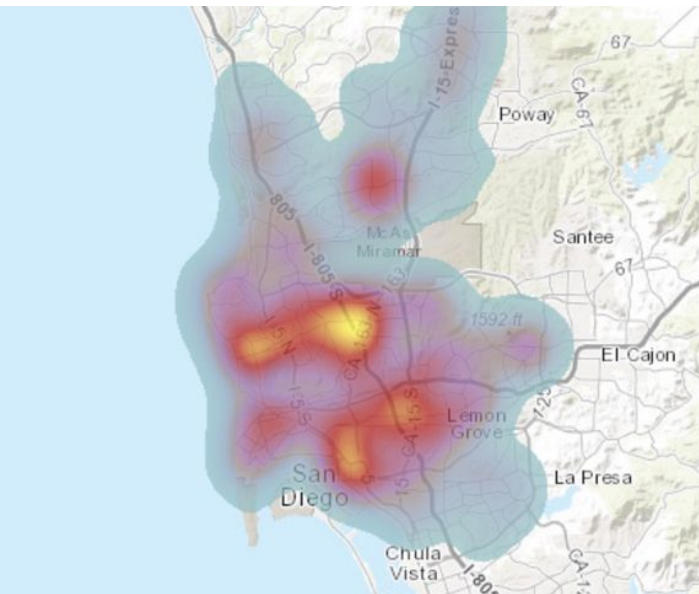
- How can we build a more realistic “null distribution” for the sample estimate without knowing the population it’s drawn from?



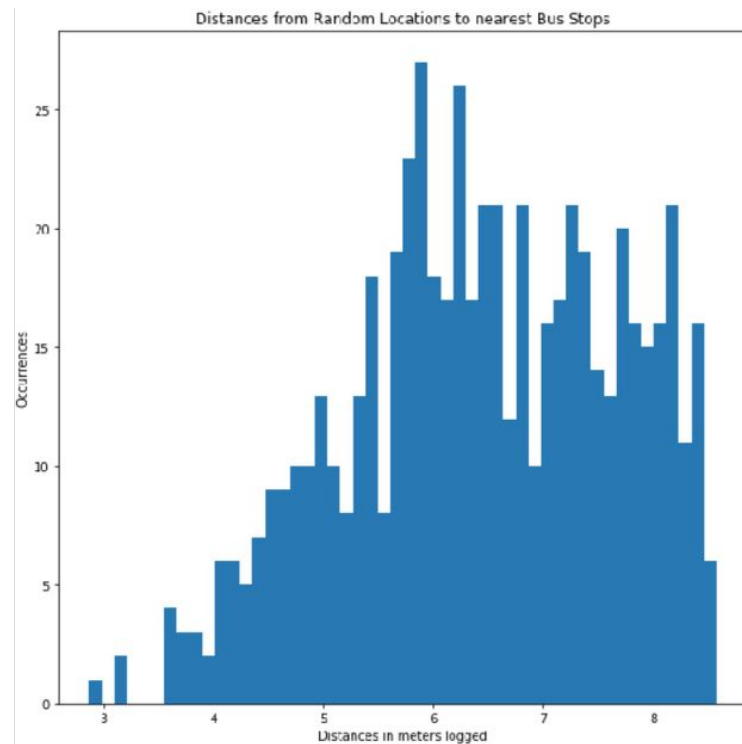
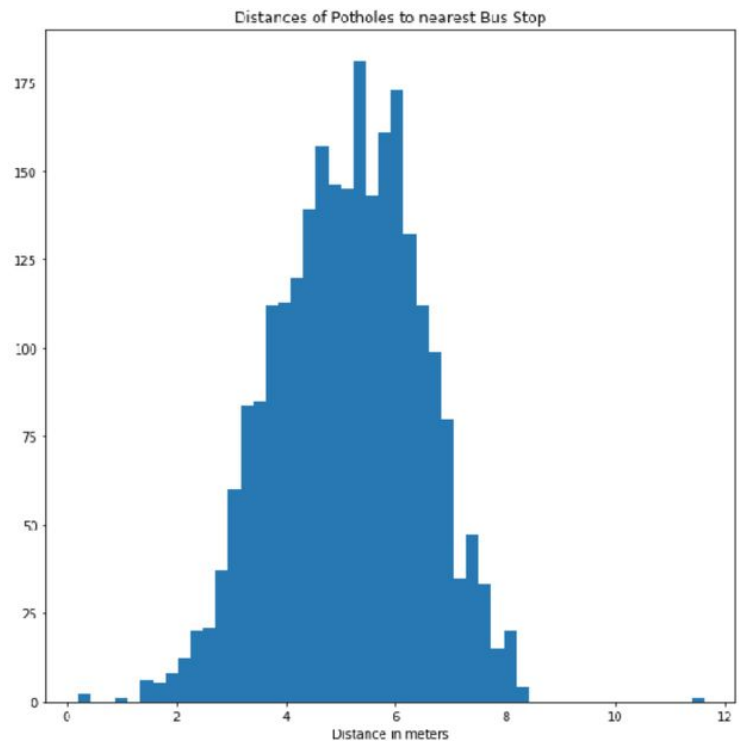
Bootstrapping (resampling)

Example Question:

- Are San Diego's pot holes closer to bus stops than not?



Bootstrapping (resampling)



2) Rank Statistics

We rank things in the real world *all the time!*

- International rankings (economics, happiness, government performance)
- Sports (teams, players, leagues)
- Search Engines
- Academic Journals' prestige
- Reviews online (1-4 stars)

Rank Statistics

Data are transformed from their quantitative value to their rank.

quantitative data

1, 4.5, 6.6, 9.2

ordinal data

1, 2, 3, 4



Ordinal data - categorical, where the variables have a natural order

Particularly helpful when data have a ranking but no clear numerical interpretation (i.e. movie reviews)

Ordinality

<https://forms.gle/jFw3rud57M8u49HCA>



Q1: Which of the following variables contains **ordinal data**?

- A horizontal blue bar with four white circular markers is positioned above the options. The options are labeled A through E.
- A Favorite Pet (i.e. dog, cat, fish, horse, etc.)
 - B Distance traveled by car each day (miles)
 - C Survey responses (scale from Dislike to Like)
 - D Human height (in inches)
 - E Human hair color (i.e. black, brown, red, blonde, etc.)

Q2: What would the **rank** of the following list be?

[77, 49, 23, 10, 89]

Wilcoxon rank-sum test (Mann Whitney U test)

- Determine whether two independent samples were selected from the same populations, having the same distribution
- Similar to t-test (but does not require normal distributions) & tests median

Assumptions:

- Observations in each group are independent of one another
- Responses are ordinal

H_0 : distributions of both populations are equal

H_a : distributions are *not* equal

Mann-Whitney U: question example

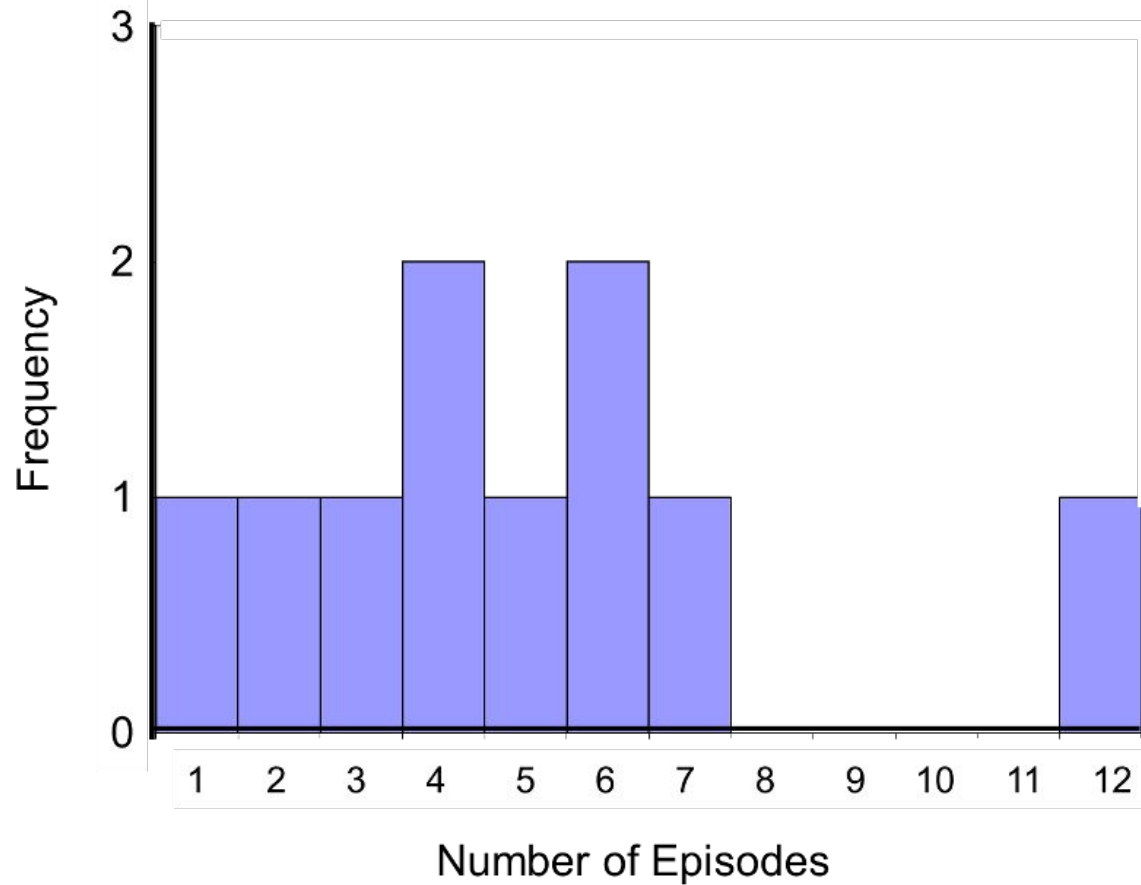
In a clinical trial, is there a difference in the number of episodes of shortness of breath between placebo and treatment?

Step 1: Participants record number of episodes they have.

Step 2: Episodes from both groups are combined, sorted, and ranked

Step 2: Resort the ranks into separate samples (placebo vs. treatment)

Step 3: Carry out statistical test



Sum of ranks:

Placebo = 37

New Drug = 18

		Total Sample (Ordered Smallest to Largest)	Ranks
Placebo	New Drug		
7	3		
5	6		
6	4		
4	2		
12	1		

Mann-Whitney U : calculating the U statistic

$$U_A = n_a n_b + \frac{n_a(n_a+1)}{2} - T_A$$

The max possible
value of T_A

The observed sum of ranks
for sample A

Ho: low and high scores are
approximately evenly distributed in
the two groups

Ha: low and high scores are NOT
evenly distributed in the two groups
($U \leq 2$)

n_a = number of elements in group A
 n_b = number of elements in group B

$$U_{\text{Placebo}} = 3$$

$$U_{\text{treatment}} = 22$$

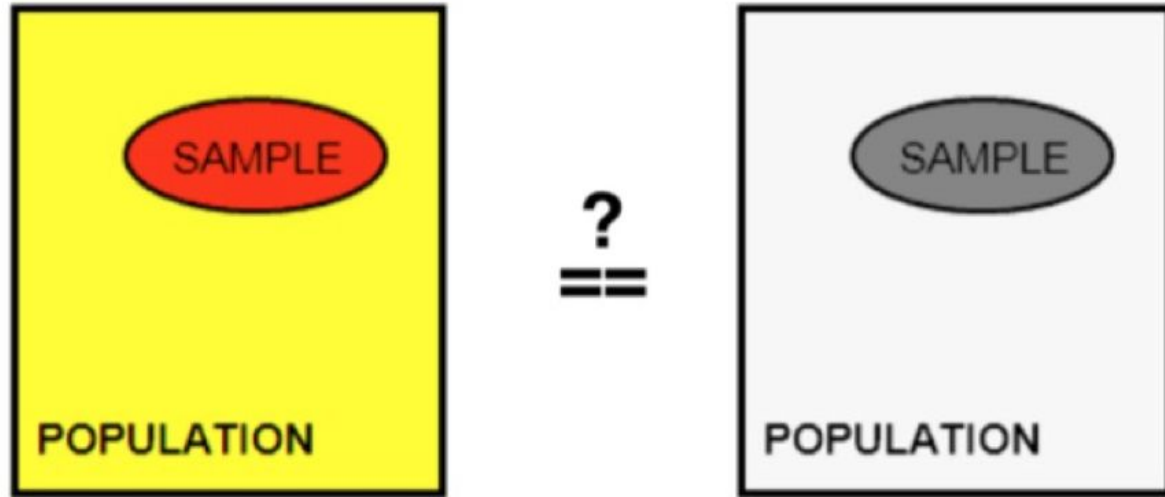
$$0 < U < n_1 * n_2$$

Complete separation $\rightarrow$ no separation

We reject the null if U is small.

3) Kolmogorov-Smirnov (KS) test

- Given (limited) samples from two populations, how do we quantify whether they come from the same distribution?

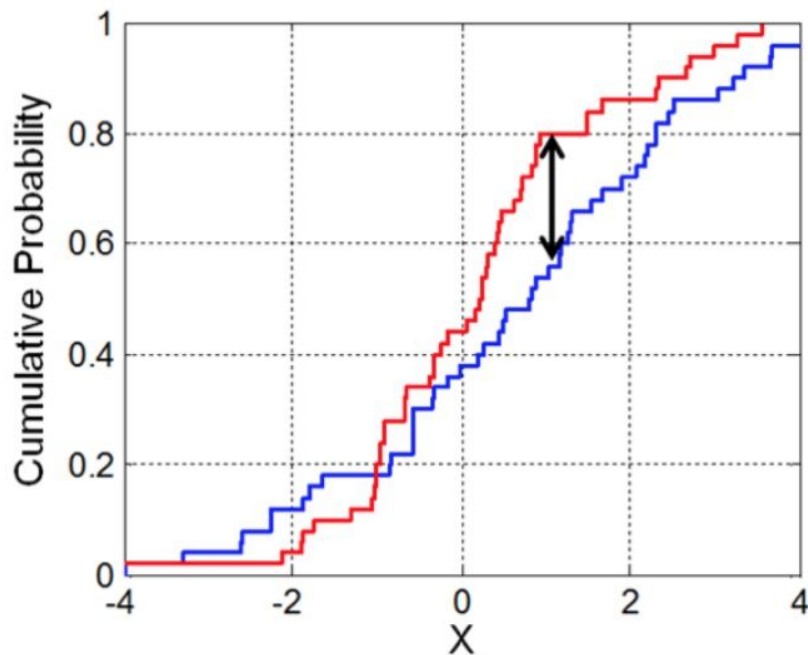


Kolmogorov-Smirnov (KS) test

Comparing cumulative distributions empirically

Tests:

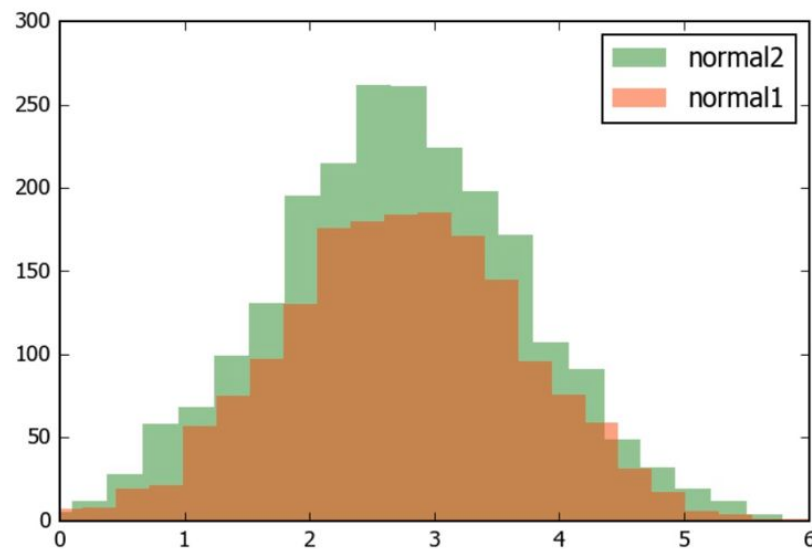
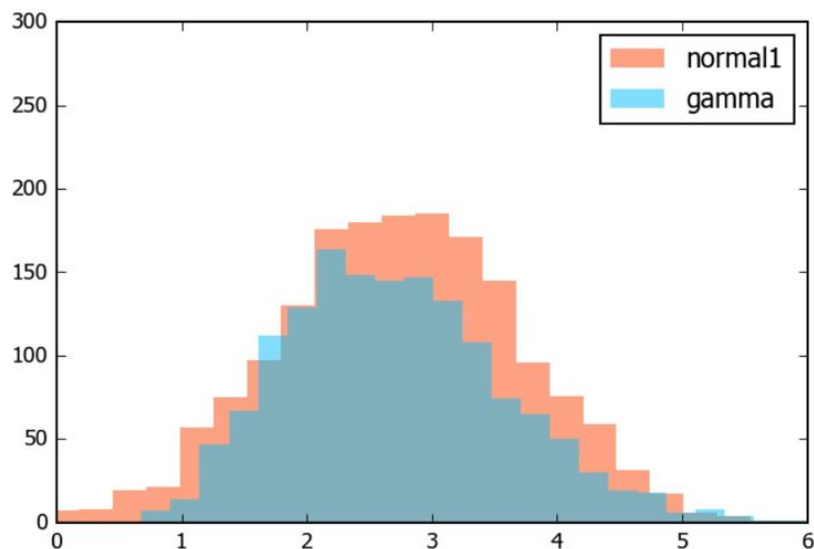
- whether a sample is drawn from a given distribution
- Whether two samples are drawn from the same distribution



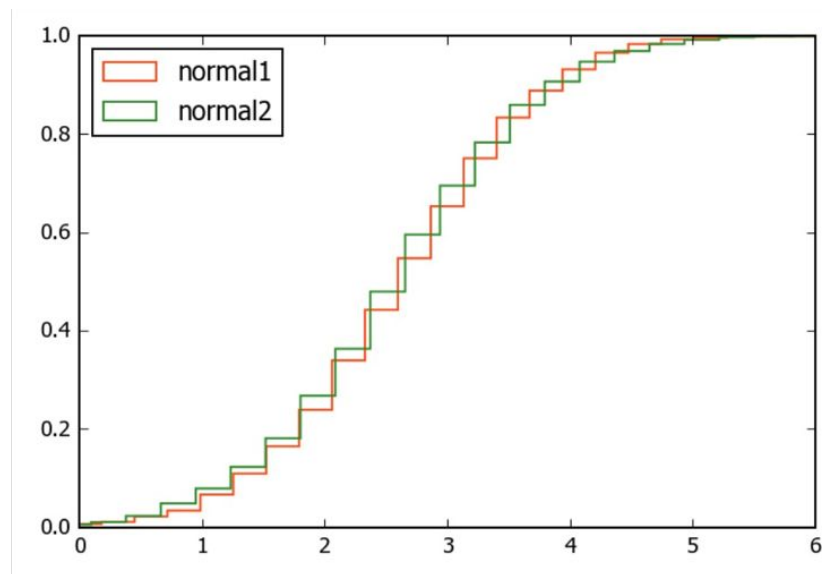
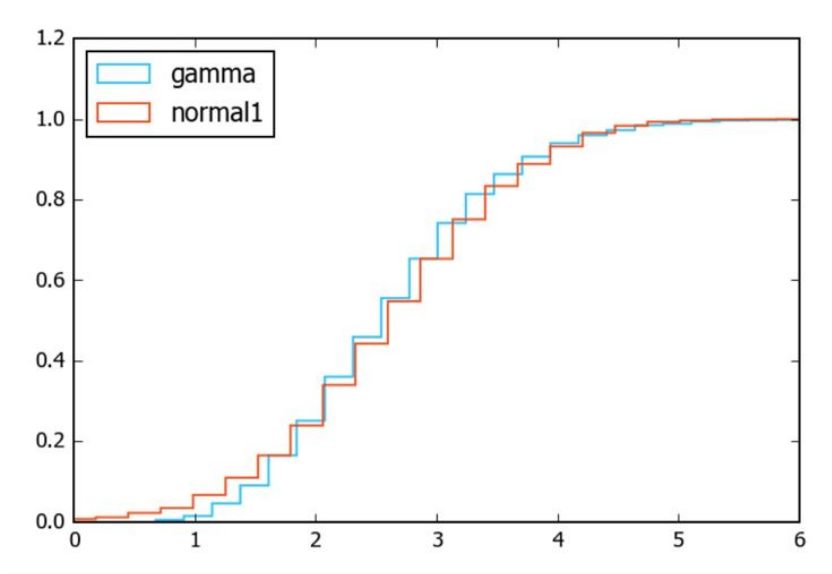
Find the maximum difference between the CDFs.

Kolmogorov-Smirnov (KS) test

- Given (limited) samples from two populations, how do we quantify whether they come from the same distribution?



Kolmogorov-Smirnov (KS) test



gamma vs. normal1: $p = 0.0106803628411$

normal1 vs. normal2: $p = 0.550735998243$

4) Non-parametric prediction models

- When you have lots of data and no prior knowledge
- When you're not focused/worried about choosing the right features
- Goal: fit training data while being able to generalize to unseen data
- Examples:
 - KNN (K-Nearest Neighbors)
 - Decision Trees & Decision Tree Regressions (CART)

Why do we even teach/use parametric statistics anyway?

Parametric approaches:

Lots of data follow expected patterns

Require less data

More sensitive

Quicker to run/train/predict

More resistant to overfitting